

Lycans AeroDesign Team
Final Project: PADA UAV Design
Section 4 (Hardware) & Section 5.2 (Software)

1 Hardware Requirements

1.1 Electrical and Avionics Design

The electrical architecture of the PADA aircraft is designed to deliver regulated, reliable power to high-draw propulsion components and sensitive digital avionics while maintaining strict weight and safety margins.

1.1.1 Component Selection and Power Budget

A comprehensive power analysis was conducted to size the power source and ensure system stability across nominal cruise and peak throttle flight regimes while meeting a strict structural weight budget.

Table 1: PADA Electrical Components Inventory

Component	Power or Amp Rating	Price	Weight	Selected Model / Notes
Battery	11.1V 3S 850mAh 75C	\$14.00	85 g	Tattu High-C Micro LiPo
ESC	30A Cont. (40A Burst)	\$12.00	18 g	BLHeli_S 30A Slim ESC
Motor	2205 2300KV (Max 22A)	\$14.00	30 g	EMAX RS2205 Outrunner
Propeller	N/A	\$3.00	6 g	Gemfan 6040 2-Blade Light
Flight Controller	5.0V DC / 2W	\$48.00	10 g	Matek H743-WING (ArduPilot)
Receiver	5.0V DC / 100mW	\$12.00	7 g	ExpressLRS EP1 Receiver
Telemetry	5.0V DC / 100mW	(included)	(shared)	ExpressLRS EP1 (combined RC + telemetry)
GPS Module	5.0V DC / 100mA	\$20.00	11 g	Matek M10Q-5883 Micro
Servos (x4)	4.8V–6.0V DC (1.8 kg · cm)	\$16.00	36 g	4x 9g Metal Gear Micro
Total System	—	\$139.00	203 g	Compliant

Note: The ExpressLRS EP1 module provides both RC control (Receiver) and telemetry over a single combined radio link; it is listed as two line items above to match the required component categories, sharing the same weight and cost entry.

Note: The 203 g electrical subsystem mass represents a self-imposed budget target reserved for avionics within the project’s 19.5 Oz (≈ 553 g) maximum gross weight constraint (Section 3), leaving adequate margin for the airframe structure.

Battery Endurance Calculation: Assuming a continuous nominal cruise draw of 9.28 A on an 850 mAh battery, total nominal operational flight time t_{flight} is calculated as:

$$t_{\text{flight}} = \frac{0.85 \text{ Ah} \times 0.80 \text{ (usable capacity)}}{9.28 \text{ A}} \times 60 \text{ min/hr} \approx 4.40 \text{ minutes} \quad (1)$$

This comfortably satisfies the 3–5 minute mission profile with a 20% capacity safety buffer while keeping avionics mass well inside the target budget. The battery energy content ($11.1\text{V} \times 0.85\text{Ah} \approx 9.4\text{Wh}$) is well within the project’s 100 Wh battery energy limit, and a LiPo (not LiHV) chemistry was selected in compliance with Section 3’s constraints.

1.1.2 Electrical Block Diagram and Schematic

Figure 1 outlines the power distribution topology and signal interconnections between the companion computer, flight controller, sensors, and power systems.

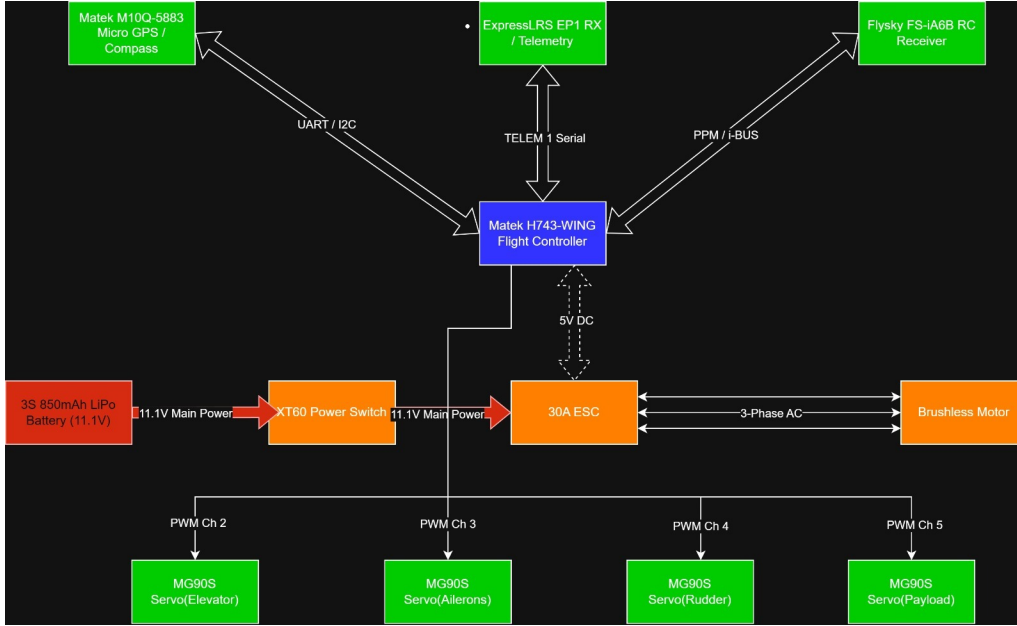


Figure 1: PADA Aircraft Electrical System Block Diagram and Circuit Interconnects.

1.1.3 Wiring, Connectors, Protection, and Electrical Safety

- **High-Current Path:** Primary power from the 3S LiPo to the ESC utilizes 16 AWG silicone wire rated for up to 30 A continuous, terminated with a lightweight XT30 anti-spark connector serving as the master physical arming switch.
- **Low-Voltage Signal Path:** Avionics and servo communication lines utilize 26 AWG twisted-pair wiring to reduce high-frequency electromagnetic interference (EMI) generated by motor switching.
- **Voltage Regulation & Protection:** Power to the Matek flight controller, sensors, and micro servos is stepped down to 5.0 V DC using the 30A ESC's integrated Battery Eliminator Circuit (BEC). An inline capacitor bank on the power distribution pads suppresses inductive voltage spikes during sudden throttle changes.

1.1.4 Component Placement and Integration

To maintain dynamic balance and aerodynamic stability, hardware components are arranged within the internal fuselage cavity as follows:

- **Center of Gravity (CG) Alignment:** The primary mass—the 3S LiPo battery (85 g) and Matek FC (10 g)—is positioned strategically over the calculated aircraft CG (25–30% mean aerodynamic chord) on an adjustable tray to allow fine pitch trimming.
- **Vibration Isolation:** The flight controller is mounted on vibration-damping foam tape to isolate the onboard IMU from high-frequency motor vibrations and maintain clean sensor fusion performance.

- **EMI Separation:** The Matek M10Q GPS/compass module is mounted externally on an elevated 10 cm carbon mast clear of motor power leads and receiver antennas to prevent electromagnetic compass offsets.

2 Software Requirements

2.1 MAVROS Installation & SITL Connection

MAVROS was installed following the official ROS 2 Jazzy package repository and connected to a running ArduPilot SITL (ArduPlane) instance over UDP. The connection was verified by monitoring the `/mavros/state` topic before proceeding to higher-level node development; a `connected: true` response confirmed a stable MAVLink link between MAVROS and the SITL instance.

2.2 Autonomous Mission Adaptation via Visual Perception

2.2.1 Physical System Setup & SITL Architecture

The system combines physical camera hardware, software perception, high-level decision-making, and Software-in-the-Loop (SITL) flight simulation:

- **Physical Optical Sensing:** A smartphone camera, connected via the IP Webcam application over a shared Wi-Fi network, serves as the live camera feed. The phone streams video over HTTP, which is captured on the companion workstation using OpenCV's VideoCapture interface.
- **Companion Workstation:** A laptop running Ubuntu 24.04 with ROS 2 Jazzy processes the live video feed. A dedicated ROS 2 node (`camera_publisher_node`) captures frames from the phone stream and publishes them on the `/camera/image_raw` topic using the `sensor_msgs/Image` message type via `cv_bridge`.
- **ROS 2 / MAVROS Middleware:** A second node (`vision_processor_node`) subscribes to the image stream, performs the perception and decision logic, and communicates with MAVROS through its service interface (`/mavros/set_mode`) to convert high-level decisions into MAVLink commands.
- **Flight Execution & Telemetry:** The MAVLink commands are received by a simulated ArduPlane instance running inside ArduPilot SITL. Flight mode changes and telemetry are monitored live in QGroundControl (QGC).

Note on Flight Control: No low-level control loops (servo control, stabilization, or speed holding) were implemented; ArduPilot handles all low-level flight control. The system's scope is strictly high-level: vision, decision-making, and mission-mode navigation.

2.2.2 Unified Mission Scenario & Perception Requirements

Rather than writing separate, unrelated scripts for obstacle avoidance and command detection, a single unified pipeline handles two real-time situations during one continuous mission flight:

4. **Environmental Hazard / Obstacle Avoidance:** The aircraft executes a baseline AUTO waypoint mission. When the camera detects a red-colored object (simulating an environmental hazard), the system switches the flight mode to GUIDED, halting the waypoint mission to avoid the hazard. Once the hazard clears from the camera's view, the system automatically reverts the flight mode to AUTO, resuming the original mission from the current waypoint.

5. **Visual Mission Instruction (Flight-Mode Command):** When the camera detects a blue-colored visual target, the autonomy layer interprets this as an explicit flight command and switches the flight mode to RTL via the `/mavros/set_mode` service call.

Table 2 documents the complete target-to-mode mapping used by the system.

Visual Cue	Interpretation	Resulting Flight Mode
Red object detected	Environmental hazard	GUIDED (evasive)
Hazard cleared (no red)	Path clear	AUTO (resume mission)
Blue object detected	Visual Target A (mission command)	RTL
No recognizable cue	Normal monitoring	No action taken

Table 2: Target-to-flight-mode mapping used by `vision_processor_node`.

2.2.3 System Architecture & ROS 2 Interface Design

The software architecture follows a modular, decoupled two-node design pattern, matching the logical flow required by the mission:

[Visual Perception] → [State Interpretation] → [Autonomy / Decision Engine] → [ROS 2 / MAVROS Gateway] → [ArduPilot SITL / QGC]

- **camera_publisher_node (Publisher):** Interfaces with the physical smartphone camera via IP Webcam, capturing frames at approximately 30 FPS, resizing them to 640×480 , and publishing them as `sensor_msgs/Image` messages over `/camera/image_raw`.

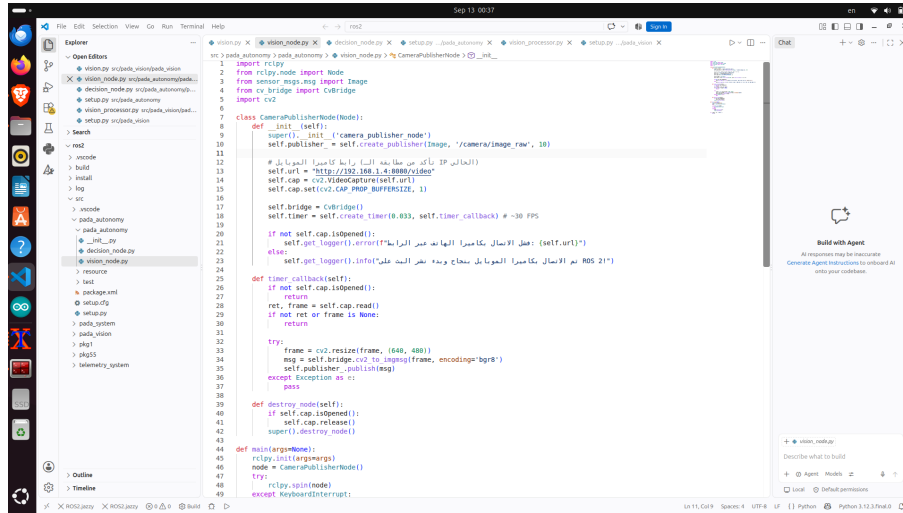


Figure 2: `camera_publisher_node` publishing camera image frames.

- **vision_processor_node (Subscriber & Service Client):** Subscribes to the image topic, processes frames using HSV color thresholding to detect hazards and visual targets, and calls the MAVROS `/mavros/set_mode` service (`mavros_msgs/SetMode`) to request flight mode changes.

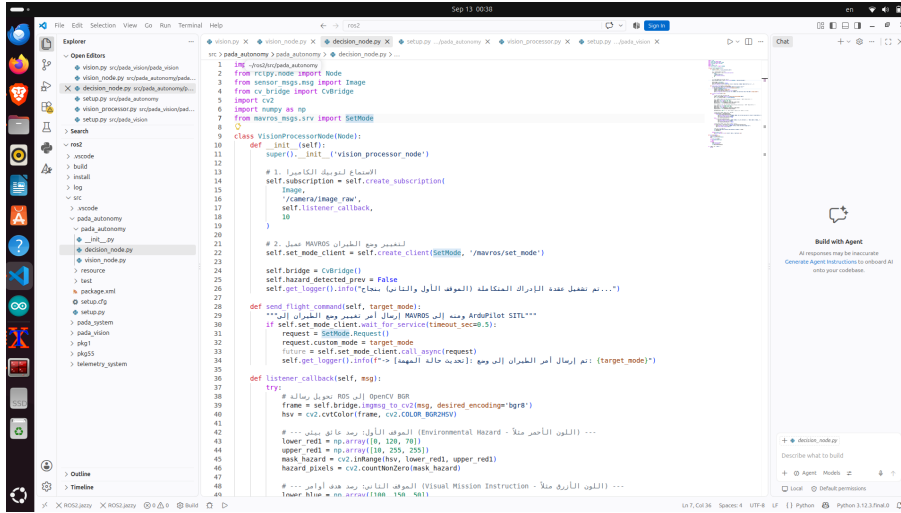


Figure 3: vision_processor_node subscribing to image topics and processing frames.

Topic/Service	Type	Purpose
/camera/image_raw	sensor_msgs/Image	Live camera feed
/mavros/set_mode	mavros_msgs/SetMode	Flight mode change requests

Table 3: ROS 2 topics and MAVROS service interfaces.

This two-node design keeps sensing and decision-making decoupled: the publisher node is solely responsible for reliable camera acquisition, while the processor node focuses entirely on perception and decision logic. This separation simplifies debugging, since each node's behavior can be verified independently by inspecting its published topic.

Error Handling: If a frame fails to be captured (`cap.read()` returns `False`), `camera_publisher_node` skips the cycle and retries on the next timer callback without crashing the node. `vision_processor_node` wraps its frame-processing logic in a `try/except` block and uses a non-blocking service wait (`wait_for_service(timeout_sec=0.5)`) so that a transient decoding error or unresponsive MAVROS service never freezes the subscription loop.

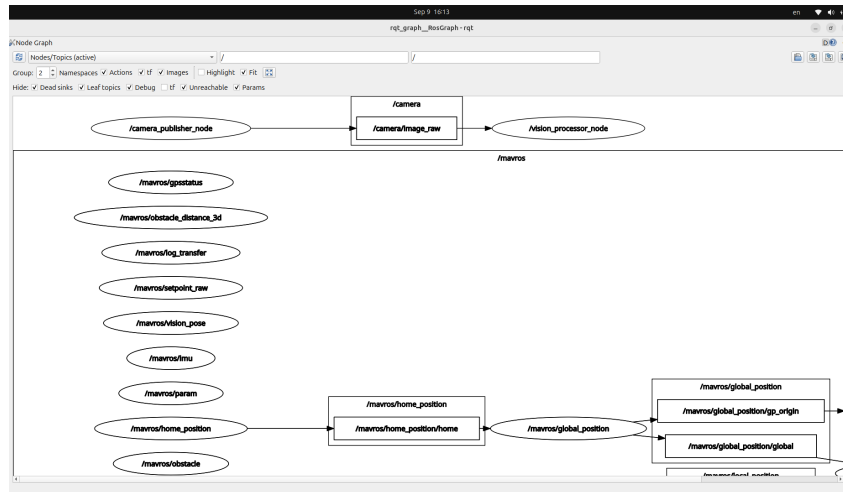


Figure 4: rqt_graph output showing the two active nodes (`camera_publisher_node`, `vision_processor_node`), the `/camera/image_raw` topic connecting them, and the MAVROS topic/service namespace used for flight mode control.

2.2.4 Engineering Justifications & Tradeoff Defense

- **Perception Strategy:** HSV color space was selected over RGB because it decouples color information (Hue) from lighting intensity (Value and Saturation), ensuring robustness against variable ambient lighting, shadows, and camera shake in dynamic environments.
- **Detection Reliability & Filtering:** To eliminate false positives and visual noise from triggering unintended flight mode changes, a strict pixel-area threshold (`threshold_area = 600 pixels`) was enforced. A state transition only occurs when the detected color mask's non-zero pixel count exceeds this threshold, preventing small color artifacts or noise from being misread as a valid hazard or target.
- **Visual Command Design:** Distinct, high-contrast colors (red for hazards/evasion, blue for operational targets) were assigned to specific flight modes (`GUIDED`, `RTL`, `AUTO`). The hazard and target logic are mutually exclusive: the target-detection check is only evaluated when no hazard is currently active, preventing conflicting simultaneous commands.
- **Obstacle Assessment & Path Planning:** At the current high-level scope of the project (per the assignment's note that no low-level control loops are required), obstacle response is implemented as an immediate mode switch to `GUIDED` upon hazard detection, which halts the `AUTO` waypoint mission and hands control to the operator/ autonomy layer. Once the hazard clears, the system autonomously reverts to `AUTO`, resuming waypoint navigation from the current mission leg without requiring mission re-upload.
- **Computational Efficiency:** By setting the video capture buffer size to 1 (`CAP_PROP_BUFFERSIZE = 1`) and resizing frames to 640×480 , the vision pipeline consistently operates at approximately 30 FPS with minimal CPU footprint, ensuring real-time responsiveness on standard workstation hardware.

2.2.5 Demonstration & Validation Protocol

The complete system was validated in a single continuous SITL flight session, demonstrating all required states:

1. **Normal Flight:** With an `AUTO` waypoint mission uploaded and active, the aircraft followed its `QGroundControl` mission plan smoothly while no visual cues were present to the camera.

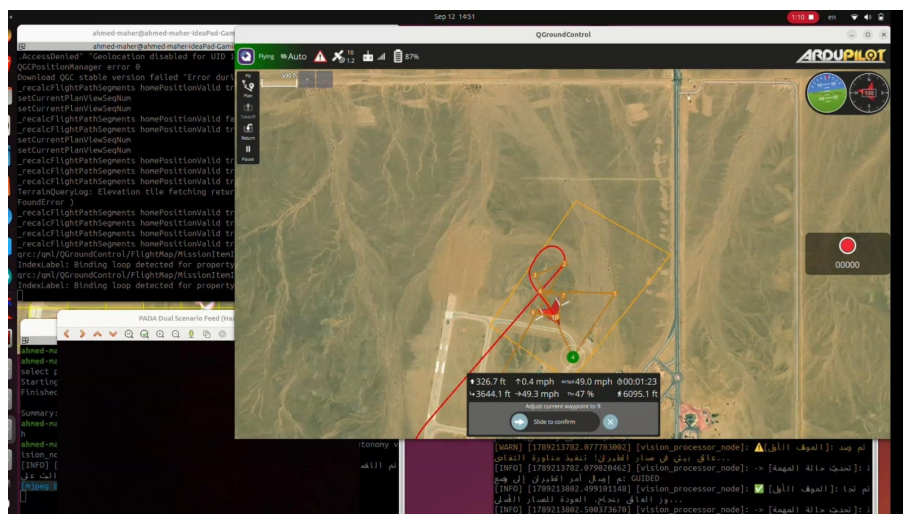


Figure 5: QGroundControl showing active `AUTO` waypoint mission execution.

2. **Hazard Avoidance & Recovery:** When a red object was presented to the camera, the system detected the hazard, switched the flight mode to GUIDED, and, once the object was removed from view, autonomously reverted to AUTO, resuming the original mission.

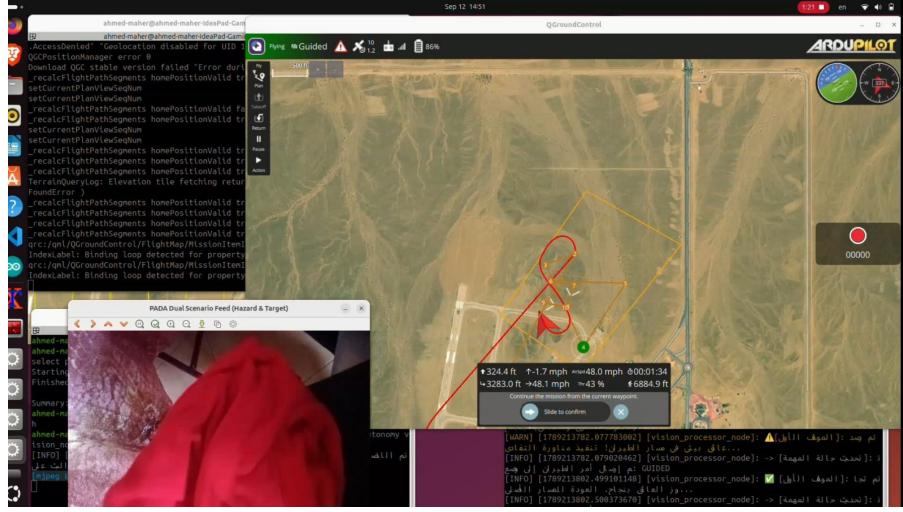


Figure 6: QGroundControl displaying transition to GUIDED mode upon hazard detection.

3. **Visual Command Processing:** When a blue object was presented to the camera, the system decoded the visual target and successfully triggered a mode change to RTL via the MAVROS /mavros/set_mode service.

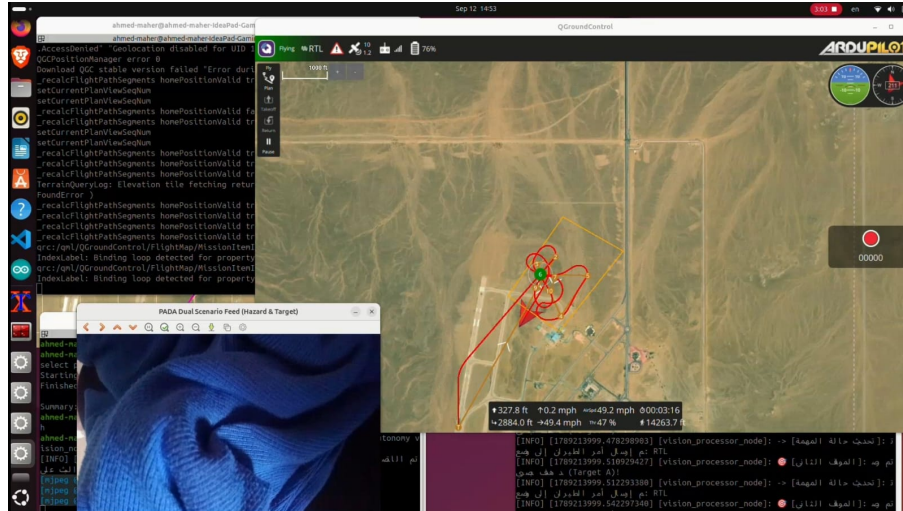


Figure 7: QGroundControl confirming flight mode switch to RTL via visual command processing.

4. **Handling Glitches & Confusion:** When the camera was pointed at an ambiguous, blurry, or partially obstructed scene (neither red nor blue exceeding the detection threshold), the system remained in its current flight mode and did not trigger any erroneous mode changes, confirming the effectiveness of the pixel-area filtering described above.